Sharp Late Coefficients and Borel Continuation of Resonant Lambert Inverses

Hyperbolic amplitudes, uniform reversion,
and the two nearest arithmetic actions

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Abstract

We resolve the explicit late-coefficient conjecture in the incoming report *Double Resonance and Nested Stokes Geometry in Polylogarithmic Lambert Inversion*. Its resonant forward tail has only one odd coefficient, whereas nonlinear inversion creates every odd sector. We prove the predicted $\cosh(\pi^2 u)$ and $\sinh(\pi^2 u)$ amplitudes by a uniform factorial-convolution estimate, give the first inverse-index correction, and retain uniformity as the slow parameter u tends to zero. A finite-phase version identifies the linear forward coefficient as the quantity that dresses the leading action amplitudes. The two Borel endpoints admit explicit logarithmic models with a continuous remainder; after the first correction, dilogarithms give a model with a continuously differentiable remainder. A separate application of Sauzin's nonlinear resurgence theorems proves continuation outside $4\pi^2\mathbb{Z}$ and summability away from the real directions. We specify which claims are new calculations, which are applications of established theory, and which questions remain open. Exact finite checks and reproducible numerical diagnostics accompany the proofs.

Research status. Conventional mathematical proofs and executed computational checks. This is an unrefereed research contribution prepared with ChatGPT assistance. Global publication priority and proof-assistant formalization are not claimed. The source snapshot and precise integration updates are recorded in Section 10.

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1 The problem and the contribution

Let $d(m)$ be the number of positive divisors of m. The resonant Lambert function

$$\mathcal{F}(t) = \sum_{m\geq 1} \frac{d(m)}{m} e^{-mt} = \sum_{a\geq 1} \frac{\mathrm{Li}_1(e^{-at})}{a}, \qquad \mathrm{Re}\, t > 0, \tag{1.1}$$

is a direct bridge between the two programs named in the research request. It is a superposition of polylogarithms, but its expansion at $t = 0$ has a factorially divergent tail. Inverting its large logarithmic core produces ordinary exponential sectors with a second, much smaller, exponential scale inside their Stokes discontinuity.

The incoming manuscript [1] obtains an exact finite formula for the inverse-sector coefficients and proves their Borel radius. Its Section 11.1, equations labelled `eq:conj-even` and `eq:conj-odd`, leaves the leading coefficient amplitudes conjectural. This article proves those formulas and their first correction. It also answers the continuation part of the next research question, Section 11.2, by checking the hypotheses of an established nonlinear resurgence theorem.

1.1 Notation and the two independent variables

Throughout, put

$$A = 4\pi^2, \qquad c = \frac{A}{4} = \pi^2, \qquad d = \frac{A^2}{288} = \frac{\pi^4}{18}, \qquad b_n = \frac{\Gamma(n)}{n} \quad (n \geq 1). \tag{1.2}$$

The lower-case constant d in (1.2) has no argument and is distinct from the divisor function $d(m)$. Define the formal series

$$R(t) = \sum_{n\geq 1} r_n t^n, \qquad r_1 = -\frac{1}{4}, \quad r_n = \begin{cases} 4b_n A^{-n}\zeta(n)^2, & n \geq 2 \text{ even}, \\ 0, & n \geq 3 \text{ odd}. \end{cases} \tag{1.3}$$

The central object is the unique formal solution

$$V(q,u) + \frac{u}{2} V(q,u)^2 + uR(qe^{-V(q,u)}) = 0, \qquad V(q,u) = \sum_{n\geq 1} C_n(u) q^n. \tag{1.4}$$

Here q and u are independent variables: q is the sector variable and u is the slow parameter. The equality is an identity of formal series, not an evaluation of the divergent series R.

To recover the physical inverse, write

$$C_* = \frac{\zeta(2)}{2} - \gamma^2 - 2\gamma_1, \quad X = \sqrt{2(Y - C_*)}, \quad u = X^{-1}, \quad q = e^{\gamma - X}, \quad t = qe^{-V}. \tag{1.5}$$

The positive square root is used for large real Y. The constant γ_1 uses the convention $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + O(s^2)$. The actual inverse of (1.1) has (1.4) as its formal sector expansion under (1.5). Large coefficient index n and large target Y are different limits; our uniform estimates make their interaction explicit.

1.2 The resolved conjecture

For $\epsilon \in \{0, 1\}$ set

$$H_0(z) = \cosh z, \qquad H_1(z) = \sinh z,$$

and let a prime denote differentiation with respect to z.

Theorem 1.1 (Sharp uniform late coefficients)**.** *For every $U > 0$, uniformly for complex $|u| \le U$ as $n \to \infty$,*

$$\frac{(-1)^{n+1}A^n C_n(u)}{4ub_n} = H_{n \bmod 2}(z) + \frac{K_{n \bmod 2}(u)}{n} + O_U(n^{-2}), \qquad z = cu, \tag{1.6}$$

$$K_\epsilon(u) = \left(du + \frac{1+u}{2}z^2\right) H_\epsilon(z) + (2+u)zH'_\epsilon(z). \tag{1.7}$$

The quotient at $u = 0$ is removable. For odd n, the error term on the right of (1.6) is in fact $O_U(|u|n^{-2})$.

In particular, uniformly for every real $0 < u \le U$,

$$C_{2m}(u) = -\frac{4u\Gamma(2m)}{2mA^{2m}} \cosh(Au/4)\{1 + O_U(m^{-1})\}, \tag{1.8}$$

$$C_{2m+1}(u) = \frac{4u\Gamma(2m+1)}{(2m+1)A^{2m+1}} \sinh(Au/4)\{1 + O_U(m^{-1})\}. \tag{1.9}$$

No lower bound on u is required.

The leading amplitudes in (1.8)–(1.9) are the precise conjecture in [1]. The estimate, its moving-parameter scope, and (1.7) strengthen that conjecture. Sections 3 and 5 supply elementary coefficient proofs. They do not rely on a Borel continuation hypothesis.

The second principal result is analytic.

Theorem 1.2 (Borel radius and continuation)**.** *For every fixed $u \in \mathbb{C} \setminus \{0\}$, the shifted Borel transform*

$$\widehat{V}_u(\xi) = \sum_{n \ge 1} \frac{C_n(u)}{\Gamma(n)} \xi^{n-1} \tag{1.10}$$

has radius exactly A, with genuine singularities at both A and $-A$. Its principal germ admits analytic continuation along every path beginning at the origin and thereafter avoiding $A\mathbb{Z}$. The formal series $V(q,u)$ is 1-summable in each direction not congruent to 0 or π modulo 2π. For $u = 0$ both V and its Borel transform vanish.

The continuation assertion is an application of Sauzin's theorems [6], after a specific change of coordinates that we prove in Section 7. It determines an allowed singularity set; it does not assert that every lattice point is singular on every sheet. Section 6 gives explicit boundary models involving logarithms and dilogarithms, with a continuously differentiable remainder.

1.3 Relationship to existing work

The forward Lambert family and its resurgent arithmetic structure are established work of Dorigoni–Kleinschmidt and Broadhurst–Dorigoni [2, 3]. The present paper studies the late coefficients of a specific nonlinear inverse. The starting coefficient formula, polynomial signs, Catalan degree, and exact forward Stokes jump belong to the incoming report [1]; we reproduce short derivations where they make this paper self-contained.

The mechanism of one large factorially weighted component has substantial antecedents. Bender–Richmond studied rapidly growing Lagrange coefficients [4]; Borinsky developed a complete algebra for factorial coefficient asymptotics, including composition and inversion [5]. In our input the two equal-modulus contributions A^{-n} and $(-A)^{-n}$ coexist. We therefore do not apply a theorem for a single positive action without checking the additional phase structure. Our proof works directly with the finite Lagrange sum and supplies the uniform

bound missing from the source conjecture. The finite-phase theorem in Section 4 records the same calculation in reusable form.

The novelty claimed here is scoped: a resolved project conjecture, explicit first corrections and Borel boundary formulas for this model, a uniform extension to a vanishing slow parameter, and a checked application of nonlinear resurgence that answers a stated continuation question. Worldwide priority for each model-specific formula has not been established.

2 The forward expansion and the exact reversion formula

2.1 How a sum of polylogarithms acquires factorial coefficients

For $t > 0$, every summand in (1.1) is positive and has negative derivative. Normal convergence on compact subsets of $\operatorname{Re} t > 0$ proves holomorphy, and monotonicity gives a positive real inverse. The Mellin representation is

$$\mathcal{F}(t) = \frac{1}{2\pi \mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} \Gamma(s)\zeta(1+s)^2 t^{-s}\,\mathrm{d}s, \qquad \sigma > 0. \tag{2.1}$$

It follows by Mellin inversion of e^{-mt} and the absolutely convergent Dirichlet identity $\sum d(m)m^{-1-s} = \zeta(1+s)^2$.

At $s = 0$ the integrand has a triple pole. Multiplication of the Laurent series of $\Gamma(s)$ and $\zeta(1+s)$ yields the residue

$$\frac{1}{2}(\log(1/t) + \gamma)^2 + C_*.$$

The residue at $s = -n$ is

$$r_n = \frac{(-1)^n}{n!}\zeta(1-n)^2. \tag{2.2}$$

The zeta functional equation gives (1.3); standard special-function conventions follow [7]. In particular $r_2 = 1/288$, and every odd coefficient after r_1 vanishes at a trivial zeta zero. Shifting the Mellin contour by any fixed number of poles gives the usual finite-order asymptotic expansion

$$\mathcal{F}(t) \sim \frac{1}{2}(\log(1/t) + \gamma)^2 + C_* + R(t), \qquad t \to 0^+. \tag{2.3}$$

Only finitely many terms are asserted at each remainder order. The factor $\Gamma(n)/n$ in (1.3) proves that R has radius zero.

Substituting (1.5) in (2.3) and cancelling $X^2/2 + C_*$ gives $XV + V^2/2 + R(qe^{-V}) = 0$, which is (1.4) after division by X. This calculation explains both the quadratic feedback and the exponential composition; neither is introduced as an arbitrary test equation.

2.2 Formal existence and a finite coefficient formula

Proposition 2.1 (Exact finite reversion)**.** *Equation* (1.4) *has a unique solution in* $q\mathbb{Q}[u][[q]]$. *For* $1 \le k \le n$, *define*

$$T_{n,k}(u) = \sum_{j=0}^{k-1} \binom{k-1}{j} n^{k-1-j} (k)_j \left(\frac{u}{2}\right)^j, \tag{2.4}$$

where $(k)_j = k(k+1)\cdots(k+j-1)$ *and* $(k)_0 = 1$. *Then*

$$C_n(u) = -\sum_{k=1}^{n} \frac{u^k}{k!}[q^n]R(q)^k\, T_{n,k}(u). \tag{2.5}$$

Proof. The coefficient of V at $q = V = 0$ is 1, so coefficient recursion gives formal existence and uniqueness over $\mathbb{Q}[u]$. To obtain the displayed formula, insert an auxiliary variable η and write

$$V = -\eta u \, \frac{R(qe^{-V})}{1 + uV/2}.$$

Formal Lagrange inversion in η gives

$$V = \sum_{k\geq 1} \frac{(-\eta u)^k}{k!} \, \partial_v^{k-1}\big(R(qe^{-v})^k (1 + uv/2)^{-k}\big)\Big|_{v=0}.$$

Only $k \leq n$ contributes to q^n, making $\eta = 1$ a legitimate coefficientwise specialization. Since $[q^n]R(qe^{-v})^k = e^{-nv}[q^n]R(q)^k$, Leibniz' rule turns the derivative into $(-1)^{k-1}T_{n,k}(u)$. This proves (2.5). □

Proposition 2.2 (Signs, valuations, and the first coefficients)**.** *For all $n \geq 1$,*

$$(-1)^{n+1} C_n(u) \in \mathbb{Q}_{\geq 0}[u] \setminus \{0\}, \tag{2.6}$$

$$\deg C_n = 2n - 1, \qquad [u^{2n-1}]C_n = \frac{(-1)^{n+1}\,\mathrm{Cat}_{n-1}}{2^{3n-1}}, \tag{2.7}$$

$$\mathrm{ord}_{u=0}\, C_n = \begin{cases} 1, & n = 1 \text{ or } n \text{ even}, \\ 2, & n \geq 3 \text{ odd}. \end{cases} \tag{2.8}$$

For odd $n \geq 3$, $[u^2]C_n = nr_{n-1}/4$. Explicitly,

$$\begin{aligned} C_1(u) &= \frac{u}{4}, \\ C_2(u) &= -\frac{u}{288} - \frac{u^2}{16} - \frac{u^3}{32}, \\ C_3(u) &= \frac{u^2}{384} + \frac{7u^3}{288} + \frac{3u^4}{128} + \frac{u^5}{128}. \end{aligned} \tag{2.9}$$

Proof. In any nonzero monomial contributing to $[q^n]R^k$, the number of linear factors has the parity of n. Hence $(-1)^n[q^n]R^k \geq 0$, and all coefficients of $T_{n,k}$ are nonnegative. The top degree comes only from $k = n$, $j = n - 1$, using $r_1^n = (-1/4)^n$; evaluation gives (2.7). The linear coefficient is $-r_n$. For odd $n \geq 3$, the quadratic coefficient is $-n[q^n]R^2/2 = -n(2r_1 r_{n-1})/2 = nr_{n-1}/4 > 0$. The displayed first polynomials follow from (2.5). These structural facts were proved in [1]; the proof is included because the exact valuations are essential to uniformity at $u = 0$. □

3 Uniform factorial convolution and the hyperbolic amplitudes

The source conjecture correctly identifies the contribution of any fixed number of factors in (2.5). The missing step is to control that number uniformly. A factorial convolution estimate supplies this control without fitting late coefficients or assuming resurgence.

3.1 A majorant that retains the endpoint factorial

Scale the coefficients by

$$a_m = A^m(-1)^m r_m, \qquad a_1 = c, \qquad a_{2j} = 4b_{2j}\zeta(2j)^2, \qquad a_{2j+1} = 0 \ (j \geq 1), \tag{3.1}$$

and set

$$S_{n,k} = [x^n]\left(\sum_{m\geq 1} a_m x^m\right)^k, \qquad D_n(u) = (-1)^{n+1}A^n C_n(u). \tag{3.2}$$

All a_m are nonnegative. If $K \geq \max\{c, 4\zeta(2)^2\}$, then $a_m \leq K b_m$ for every m. Equation (2.5) becomes

$$D_n(u) = \sum_{k=1}^{n} \frac{u^k}{k!} S_{n,k} T_{n,k}(u). \tag{3.3}$$

Lemma 3.1 (Endpoint-preserving factorial bounds)**.** *For $n \geq 2$,*

$$\sum_{m=1}^{n-1} b_m b_{n-m} \leq 3 b_{n-1}. \tag{3.4}$$

Consequently, for $1 \leq k \leq n$,

$$S_{n,k} \leq K^k 3^{k-1} b_{n-k+1}, \qquad \frac{n^{k-1} b_{n-k+1}}{b_n} \leq k e^k. \tag{3.5}$$

Proof. The ratio of the mth summand in (3.4) to b_{n-1} is

$$\frac{n-1}{m(n-m)}\binom{n-2}{m-1}^{-1} \leq \binom{n-2}{m-1}^{-1}.$$

For $N \geq 2$ every interior binomial coefficient $\binom{N}{j}$ is at least N, so $\sum_{j=0}^{N}\binom{N}{j}^{-1} \leq 2 + (N-1)/N < 3$. The cases $N = 0, 1$ satisfy the same bound. This proves (3.4). Induction on k, shifting the convolution index at each step, gives $[x^n](\sum_{m\geq 1} b_m x^m)^k \leq 3^{k-1} b_{n-k+1}$, and therefore the first part of (3.5).

For the second part let $h = k - 1$. Direct factorial cancellation gives

$$\frac{n^h b_{n-h}}{b_n} = \frac{n}{n-h}\prod_{i=1}^{h}\left(1 - \frac{i}{n}\right)^{-1}. \tag{3.6}$$

The first factor is at most k, since $(k-1)(n-k) \geq 0$. For the increasing function $f(x) = -\log(1-x)$,

$$\sum_{i=1}^{k-1} f(i/n) \leq n\int_0^{k/n} f(x)\,\mathrm{d}x = k + (n-k)\log(1-k/n) \leq k.$$

At $k = n$ the integral is finite and $0\log 0$ is interpreted as zero. Exponentiation proves the bound, including the endpoint $k = n$. □

For $|u| \leq U$, the exact polynomial (2.4) satisfies

$$|T_{n,k}(u)| \leq n^{k-1}(1+U)^{k-1}, \tag{3.7}$$

because $(k)_j \leq (2n)^j$. Combining this with Lemma 3.1 gives the central estimate

$$\frac{1}{b_n}\left|\frac{u^k}{k!} S_{n,k} T_{n,k}(u)\right| \leq \frac{k}{k!} K^k 3^{k-1} e^k U^k (1+U)^{k-1} \leq \frac{k B_U^k}{k!} \tag{3.8}$$

for a suitable fixed B_U. The final majorant is summable in k and independent of n. It removes the spurious $e^{O(\sqrt{n})}$ loss of the coarser composition count in the predecessor.

The same conclusion holds after division by u for even n, and by u^2 for odd $n \geq 3$. In the latter case the exact sum begins at $k = 2$, by (2.8). On a compact disk one uses $|u|^{k-v} \leq \max(1, U)^k$, with $v = 1$ or 2. Every quotient is a polynomial, so no estimate divides by a small nonzero number.

3.2 The contribution of one large index

Lemma 3.2 (First endpoint extraction). *For $k \geq 2$ and $n \geq k+2$,*

$$S_{n,k} = kc^{k-1}a_{n-k+1} + E_{n,k}, \qquad 0 \leq E_{n,k} \leq 6\binom{k}{2}K^k 3^{k-2} b_{n-k}. \tag{3.9}$$

For $k = 1$ the main term is exact.

Proof. Compositions with exactly one part greater than 1 give the main term. Every remaining composition has two parts at least 2. Mark a pair of such positions and use a union bound over $\binom{k}{2}$ choices.

For $N \geq 4$,

$$\sum_{m=2}^{N-2} b_m b_{N-m} \leq 6b_{N-2}. \tag{3.10}$$

Indeed, put $M = N-4$ and $j = m-2$. The ratio of a summand to b_{N-2} is

$$\frac{M+2}{M+1}\frac{(j+1)(M-j+1)}{(j+2)(M-j+2)}\binom{M}{j}^{-1} \leq 2\binom{M}{j}^{-1}.$$

The reciprocal-binomial bound proves (3.10). After summing the two marked parts, convolve with the remaining $k-2$ parts and enlarge the index range. The result is bounded by

$$6K^k[x^{n-2}]\left(\sum_{m\geq 1} b_m x^m\right)^{k-1} \leq 6K^k 3^{k-2} b_{n-k}.$$

Summing over the marked pair proves (3.9). □

For fixed k, multiplication of the remainder in (3.9) by n^{k-1}/b_n gives $O_k(n^{-1})$. The main index is $m = n-k+1$. If m is even, then $n^{k-1}a_m/b_n \to 4$; if m is odd it is eventually zero. Also $T_{n,k}(u)/n^{k-1} \to 1$ locally uniformly in u. Thus the limit of the kth summand in (3.3), divided by b_n, is

$$\frac{4ku^k c^{k-1}}{k!} = 4u\frac{(cu)^{k-1}}{(k-1)!} \tag{3.11}$$

when $n-k+1$ is even, and zero in the other parity. The majorant (3.8) now justifies summation of these limits. Even n selects even $k-1$ and produces $\cosh(cu)$; odd n selects odd $k-1$ and produces $\sinh(cu)$.

3.3 Uniform errors, including the zero-parameter boundary

Theorem 3.3 (A quantified proof of the source conjecture). *For every $U > 0$, uniformly for complex $|u| \leq U$,*

$$\frac{A^n C_n(u)}{4b_n u} = -\cosh(cu) + O_U(n^{-1}) \qquad (n \text{ even}), \tag{3.12}$$

$$\frac{A^n C_n(u)}{4b_n u^2} = \frac{\sinh(cu)}{u} + O_U(n^{-1}) \qquad (n \geq 3 \text{ odd}). \tag{3.13}$$

All quotients at $u = 0$ are interpreted by continuity.

Proof. Split the finite sum at $k = n/2$ and first take $k \le n/2$. Let $h = k - 1$ and $R_{n,k} = n^h b_{n-h}/b_n$. The logarithm of (3.6) and $-\log(1-x) \le 2x$ on $[0, 1/2]$ give

$$\begin{aligned} 0 \le \log R_{n,k} &\le \frac{h(h+3)}{n}, \\ |R_{n,k} - 1| &\le \frac{ke^k h(h+3)}{n}, \\ \frac{n^h b_{n-h-1}}{b_n} &= R_{n,k}\frac{n-h}{(n-h-1)^2} \le \frac{4ke^k}{n} \quad (n \ge 4). \end{aligned} \tag{3.14}$$

The last estimate controls (3.9). Using the sharper $(k)_j \le (2k)^j$ in (2.4) gives

$$\left|\frac{T_{n,k}(u)}{n^h} - 1\right| \le (1 + kU/n)^h - 1 \le \frac{k^2 U}{n}(1+U)^k. \tag{3.15}$$

For the even endpoint $m = n - k + 1 \ge n/2$, the error $\zeta(m)^2 - 1 = O(2^{-m})$ is uniformly $O(n^{-1})$. Combining these estimates bounds the difference between each normalized summand and (3.11) by

$$\frac{1}{n}\frac{P(k)\widetilde{B}_U^k}{k!}$$

for a fixed polynomial P. The bound also holds after division by the valuation factor u or u^2, as explained after (3.8). It is summable over k.

For $k > n/2$ both the exact summands and the limiting exponential summands have a majorant of the form $kB_U^k/k!$. The inequality $1 \le 2k/n$ makes the summed tail $O_U(n^{-1})$. More generally, using $1 \le (2k/n)^M$ makes this tail $O_{U,M}(n^{-M})$ for any fixed M. Summing (3.11) with its parity restriction proves (3.12)–(3.13). □

For real $0 \le u \le U$, one has $\cosh(cu) \ge 1$ and $\sinh(cu)/u \ge c$, including the limiting value at zero. Dividing by these lower bounds proves the uniform relative versions (1.8)–(1.9). This is stronger than an estimate whose constant depends on a fixed lower bound for u.

Corollary 3.4 (Two limits may be taken together)**.** *Let $u_n > 0$, $u_n \to 0$, and $n \to \infty$ through the indicated parity. Then*

$$C_n(u_n) \sim \begin{cases} -4u_n b_n A^{-n}, & n \text{ even}, \\ Au_n^2 b_n A^{-n}, & n \text{ odd}. \end{cases} \tag{3.16}$$

The rate at which u_n approaches zero is unrestricted.

Proof. Apply the uniform relative bounds and use $\cosh(cu) = 1 + O(u^2)$ and $\sinh(cu) = cu\{1 + O(u^2)\}$. □

The coefficient theorem therefore applies to the physical slow parameter $u = 1/X$ while the sector index increases. It does not assert that the divergent infinite sector series may be summed ordinarily at that physical pair, nor does a coefficient estimate alone determine an optimally truncated remainder of the actual inverse.

4 Finite phases and changes of the initial coefficients

The hyperbolic functions reflect the two phases 1 and -1 in the forward coefficients. The same proof yields a finite-phase transformation law, with no positivity requirement.

Theorem 4.1 (Leading amplitudes for finitely many phases)**.** *Let $A > 0$, $b_n = \Gamma(n)/n$, and $\widetilde{R}(t) = \sum_{n\geq 1} \widetilde{r}_n t^n$, with $\widetilde{r}_1 = \rho \in \mathbb{C}$. Suppose that for a finite set $E \subset \{\omega \in \mathbb{C} : |\omega| = 1\}$ of distinct phases,*

$$|\widetilde{r}_n| \leq K A^{-n} b_n, \qquad \frac{A^n \widetilde{r}_n}{b_n} = \sum_{\omega \in E} \kappa_\omega \omega^n + O(n^{-1}). \tag{4.1}$$

Let $\widetilde{C}_n(u)$ be the coefficients of the zero-constant formal solution of (1.4) with R replaced by $\widetilde{R}$. Then, locally uniformly for $u \in \mathbb{C}$,

$$\widetilde{C}_n(u) = -uA^{-n}b_n \left\{ \sum_{\omega \in E} \kappa_\omega \omega^n e^{A\rho u/\omega} + O_U(n^{-1}) \right\}. \tag{4.2}$$

The bracketed formulation extends through $u = 0$. Its error corresponds to an absolute $O_U(|u|A^{-n}b_n/n)$ error in $\widetilde{C}_n$.

Proof. Apply the exact formula (2.5) to the new input and scale by A^n. The first endpoint is now $k(A\rho)^{k-1}(A^{n-k+1}\widetilde{r}_{n-k+1})$. All remainder estimates of Lemmas 3.1 and 3.2 apply to absolute values: the positivity assumption was used to construct a majorant, not to establish an identity. For $k \leq n/2$, the remainder in (4.1) is $O(n^{-1})$ at the endpoint. The same proof as Theorem 3.3 therefore justifies the uniform k summation with error $O_U(n^{-1})$. For each phase the limiting sum is

$$-\sum_{k\geq 1} \frac{u^k}{(k-1)!} (A\rho)^{k-1} \kappa_\omega \omega^{n-k+1} = -u\kappa_\omega \omega^n e^{A\rho u/\omega}.$$

The sum over E is finite, which proves (4.2). □

The action corresponding to ω is $\lambda_\omega = A/\omega$. Its leading amplitude is multiplied by $e^{\lambda_\omega \rho u}$. This multiplier comes entirely from repeated linear factors of the forward series. The terms of $T_{n,k}$ with $j \geq 1$, which encode the quadratic part of the inverse core, enter one inverse-index order later. Cancellation between distinct phases can occur on subsequences, so (4.2) is an additive expansion. A relative equivalent requires a separate nonvanishing argument.

Corollary 4.2 (Polynomial changes of the forward tail)**.** *For the Lambert input, replace $R(t)$ by $R(t) + P(t)$, where $P(0) = 0$ is a fixed polynomial and $p_1 = [t]P(t)$. The late coefficient formula is*

$$\widetilde{C}_n(u) = -2uA^{-n}b_n \left\{ e^{A(p_1 - 1/4)u} + (-1)^n e^{-A(p_1-1/4)u} + O_U(n^{-1}) \right\}. \tag{4.3}$$

In particular, changes satisfying $P(t) = O(t^2)$ leave both leading action amplitudes unchanged.

Proof. A polynomial changes no sufficiently late forward coefficient. Apply Theorem 4.1 with $E = \{1, -1\}$ and $\kappa_1 = \kappa_{-1} = 2$. □

This corollary gives a useful separation of data: the arithmetic tail locates the leading actions, the linear term changes their amplitudes, and the quadratic term first appears in the correction computed next.

5 The first inverse-index correction

The leading amplitude only sees one large forward index and the linear indices accompanying it. At the next order, a single quadratic forward index also contributes. We retain that contribution uniformly in the number of factors, then compute the corrections from the gamma ratios and from $T_{n,k}$. This proves the full statement of Theorem 1.1.

5.1 Two endpoint terms with a uniform remainder

Recall the scaled coefficients and convolution coefficients from Section 3:

$$a_m = A^m(-1)^m r_m, \qquad S_{n,k} = [x^n]\left(\sum_{m\ge1} a_m x^m\right)^k.$$

In particular $a_1 = c$, $a_2 = d$, and every a_m is nonnegative. It is convenient to shift the index once:

$$\alpha_j = a_{j+1}, \qquad \alpha(x) = \sum_{j\ge0}\alpha_j x^j, \qquad S_{n,k} = [x^{n-k}]\alpha(x)^k. \tag{5.1}$$

Thus $\alpha_0 = c$, $\alpha_1 = d$, and there is a constant B such that

$$0 \le \alpha_j \le B\Gamma(j) \qquad (j \ge 1). \tag{5.2}$$

Indeed $a_{j+1} = 4\Gamma(j+1)\zeta(j+1)^2/(j+1)$ when $j+1$ is even, and it is zero otherwise. The ratio $j/(j+1)$ is at most one, and $\zeta(j+1) \le \zeta(2)$.

Lemma 5.1 (The second endpoint contribution)**.** *There is a constant $B_* > 0$ such that, whenever $n \ge 12$ and $1 \le k \le n/2$,*

$$\begin{aligned} S_{n,k} = {} & kc^{k-1}a_{n-k+1} + k(k-1)c^{k-2}d\,a_{n-k} \\ & + O\big(B_*^k\Gamma(n-k-2)\big). \end{aligned} \tag{5.3}$$

The second term is understood to be zero for $k = 1$, and the implied constant is independent of both n and k.

Proof. We first record two elementary estimates. The sequence $F_j = j!$ satisfies $F * F \le 3F$ coefficientwise. For $M \ge 1$, division of its Mth convolution coefficient by $M!$ gives

$$\sum_{j=0}^{M}\binom{M}{j}^{-1} \le 3 :$$

the endpoints contribute two, and each of the $M-1$ interior terms is at most $1/M$. The case $M = 0$ is immediate. Iteration, followed by the change $j_i \mapsto j_i - 1$, gives

$$\sum_{\substack{j_1+\cdots+j_h=N\\ j_i\ge1}}\prod_{i=1}^{h}\Gamma(j_i) \le 3^{h-1}\Gamma(N-h+1), \qquad N \ge h \ge 1. \tag{5.4}$$

Removing the endpoints from the two-factor convolution gives a sharper estimate:

$$\sum_{j=2}^{N-2}\Gamma(j)\Gamma(N-j) \le 4\Gamma(N-2), \qquad N \ge 4. \tag{5.5}$$

For completeness, put $M = N-2$ and divide the left side by $M!$. The result is $\sum_{i=1}^{M-1}\binom{M}{i}^{-1}$. For $M \ge 3$, its two extreme terms contribute $2/M$, and each of the remaining $M-3$ terms is at most $2/[M(M-1)]$. The sum is at most $4/M$. The case $M = 2$ also satisfies this bound, proving (5.5).

Now put $N = n - k$, so $N \geq k$ and $N \geq 6$, and expand $\alpha(x)^k$ according to the number h of positive indices. The case $h = 1$ contributes $kc^{k-1}\alpha_N$. For $h = 2$, the indices $1, N-1$ and $N-1, 1$ contribute

$$2\binom{k}{2}c^{k-2}\alpha_1\alpha_{N-1} = k(k-1)c^{k-2}d\alpha_{N-1}.$$

By (5.2) and (5.5), the remaining $h = 2$ terms are bounded by $4\binom{k}{2}c^{k-2}B^2\Gamma(N-2)$. For $h \geq 3$, (5.4) bounds the contribution by

$$\binom{k}{h}c^{k-h}B^h 3^{h-1}\Gamma(N-h+1).$$

Since $N \geq k \geq h$, one has $\Gamma(N-h+1) \leq \Gamma(N-2)$. The binomial theorem therefore bounds the sum of all these remainders by $B_*^k\Gamma(N-2)$ after a fixed increase of B_*. Substituting $\alpha_N = a_{n-k+1}$ and $\alpha_{N-1} = a_{n-k}$ proves (5.3). □

5.2 The two elementary expansions

The next estimates are uniform over the full range $k \leq n/2$; their constants may grow exponentially in k, because the Lagrange factor $1/k!$ makes that growth summable. Define

$$Q_{n,k}(u) = n^{1-k}T_{n,k}(u), \qquad \beta_k = \frac{(k-1)(k+2)}{2}.$$

For $|u| \leq U$, the finite sum (2.4) gives

$$Q_{n,k}(u) = 1 + \frac{k(k-1)u}{2n} + O_U\left(\frac{(1+k)^4 e^{B_U k}}{n^2}\right). \tag{5.6}$$

To justify the uniform remainder, use $\binom{k-1}{j} \leq k^j/j!$ and $(k)_j \leq (2k)^j$. The sum of the absolute values of the terms with $j \geq 2$ is at most

$$\sum_{j\geq 2}\frac{(Uk^2/n)^j}{j!} \leq \frac{U^2k^4}{2n^2}e^{Uk^2/n}.$$

The inequality $k^2/n \leq k/2$ proves (5.6). It also gives a bound $|Q_{n,k}(u)| \leq e^{Uk/2}$ that will be used for the endpoint remainder.

For the first endpoint, put $p = n - k + 1$. When p is even,

$$\frac{n^{k-1}a_p}{b_n} = 4\left\{1 + \frac{\beta_k}{n} + O\left(\frac{(1+k)^4 e^{B_1 k}}{n^2}\right)\right\}; \tag{5.7}$$

when p is odd it is zero. The exact gamma ratio before multiplication by $4\zeta(p)^2$ is

$$\frac{n^{k-1}b_{n-k+1}}{b_n} = \frac{n^k}{(n-k+1)\prod_{j=1}^{k-1}(n-j)}. \tag{5.8}$$

Its logarithm equals $\beta_k/n + O(k^3/n^2)$: the offsets in the denominator are $k-1, 1, \ldots, k-1$, whose sum is β_k, and every offset divided by n is at most $1/2$. More explicitly, $-\log(1-v) = v + O(v^2)$ uniformly for $0 \leq v \leq 1/2$. The logarithm itself is $O(k^2/n) = O(k)$, so exponentiating introduces a remainder $O((1+k)^4 e^{B_1 k}/n^2)$. Finally $\zeta(p)^2 = 1 + O(2^{-p})$, with $p \geq n/2$, and this exponentially small error is absorbed by the displayed bound. This proves (5.7).

The second endpoint has the corresponding estimate

$$\frac{n^{k-1}a_{n-k}}{b_n} = \frac{4}{n} + O\left(\frac{(1+k)^2 e^{B_2 k}}{n^2}\right) \quad \text{if } n-k \text{ is even,} \tag{5.9}$$

and it is zero otherwise. Here the exact gamma ratio is

$$\frac{n^{k-1}b_{n-k}}{b_n} = \frac{n^k}{(n-k)\prod_{j=1}^{k}(n-j)}.$$

Factor out $1/n$ and apply the same logarithmic estimate to the offsets $k, 1, \dots, k$. This proves (5.9).

The remainder in Lemma 5.1 is also uniformly of second order after normalization. For $n \geq 12$ and $k \leq n/2$,

$$\frac{n^{k-1}\Gamma(n-k-2)}{b_n} = \frac{n^k\Gamma(n-k-2)}{\Gamma(n)} \leq \frac{3^{k+2}}{n^2}. \tag{5.10}$$

Indeed, the denominator ratio contains $k+2$ factors $n-j$, each at least $n-k-2 \geq n/3$. Combining (5.3)–(5.10), all errors in the kth summand of (2.5), after division by b_n and scaling by A^n, are bounded by

$$\frac{|u|}{n^2}\frac{M_U^k(1+k)^M}{k!}, \tag{5.11}$$

for constants M_U and a fixed integer M. One obtains this bound by leaving one factor $|u|$ outside and replacing the remaining $|u|^{k-1}$ by U^{k-1}. Thus the estimates remain uniform after division by u, with the removable value at zero.

5.3 Summing the corrections

We now complete the proof of Theorem 1.1. Recall that $D_n(u) = (-1)^{n+1}A^n C_n(u)$. The range $k > n/2$ contributes less than any prescribed inverse power of n after normalization. To make the uniformity at $u = 0$ explicit, the estimate (3.8), with one factor u left outside, gives

$$\left|\frac{u^k S_{n,k}T_{n,k}(u)}{k!b_n}\right| \leq |u|L_U\frac{kL_U^{k-1}}{k!}, \qquad 1 \leq k \leq n. \tag{5.12}$$

It follows equally by inserting the three bounds in Lemma 3.1 into the left side. For every fixed integer $J \geq 1$, use $1 \leq (2k/n)^J$ when $k > n/2$ to obtain

$$\sum_{k>n/2}\left|\frac{u^k S_{n,k}T_{n,k}(u)}{k!b_n}\right| \leq \frac{|u|\,2^J L_U}{n^J}\sum_{k\geq 1}\frac{k^{J+1}L_U^{k-1}}{k!} = O_{U,J}(|u|n^{-J}).$$

The tails of the exponential series below, including their polynomial weights in k, satisfy the same estimate. This permits replacement of the remaining finite sums by infinite sums at the required error order.

For $k \leq n/2$, the first endpoint in (5.3) contributes only when $n-k+1$ is even. Put $\ell = k-1$; its parity condition becomes $\ell \equiv n \pmod 2$. After division by $4ub_n$, its contribution is

$$\frac{z^\ell}{\ell!}\left[1 + \frac{\ell(\ell+3) + u\ell(\ell+1)}{2n}\right], \qquad z = cu,$$

up to the summable error (5.11) divided by $4|u|$. For the second endpoint put $\ell = k-2$. Its parity condition is again $\ell \equiv n \pmod 2$, and its first-order contribution is $du\, z^\ell/(n\ell!)$. Hence

$$\begin{aligned}\frac{D_n(u)}{4ub_n} = &\sum_{\substack{\ell\ge 0\\ \ell\equiv n\ (2)}} \frac{z^\ell}{\ell!}\\ &+\frac{1}{n}\sum_{\substack{\ell\ge 0\\ \ell\equiv n\ (2)}} \left\{du + \frac{\ell(\ell+3)+u\ell(\ell+1)}{2}\right\}\frac{z^\ell}{\ell!} + O_U(n^{-2}).\end{aligned} \tag{5.13}$$

All sums and remainders are locally uniform in complex u.

If $H = H_{n \bmod 2}$, the elementary identities

$$\sum_{\ell\equiv n\ (2)} \ell(\ell+3)\frac{z^\ell}{\ell!} = z^2H(z) + 4zH'(z), \qquad \sum_{\ell\equiv n\ (2)} \ell(\ell+1)\frac{z^\ell}{\ell!} = z^2H(z) + 2zH'(z)$$

turn (5.13) into (1.6)–(1.7).

It remains to prove the extra factor u in the odd remainder. For odd $n \ge 3$, let

$$E_n(u) = \frac{D_n(u)}{4ub_n} - H_1(cu) - \frac{K_1(u)}{n}.$$

This is entire in u. By the exact valuation in (2.8), and by the displayed formulas for H_1 and K_1, it vanishes at zero. The already proved estimate on the larger disk $|u| \le 2U$ bounds E_n there by $O_{2U}(n^{-2})$. Apply the maximum-modulus principle to the entire function $E_n(u)/u$ on that disk: its boundary values are $O_{2U}(n^{-2}/U)$. Consequently

$$E_n(u) = O_U(|u|n^{-2}), \qquad |u| \le U. \tag{5.14}$$

This completes the complex-uniform part of Theorem 1.1.

Corollary 5.2 (Relative uniformity down to the resonant endpoint)**.** *For every fixed $U > 0$, uniformly for real $0 < u \le U$,*

$$C_n(u) = (-1)^{n+1}\frac{4ub_n}{A^n}H_{n \bmod 2}(cu)\left\{1 + \frac{K_{n \bmod 2}(u)}{nH_{n \bmod 2}(cu)} + O_U(n^{-2})\right\}. \tag{5.15}$$

The ratios $K_0(u)/H_0(cu)$ and $K_1(u)/H_1(cu)$ extend continuously to $u = 0$, with respective limits 0 and 2.

Proof. For $u \ge 0$, $\cosh(cu) \ge 1$ and $\sinh(cu) \ge cu$. Divide the even remainder by the former bound and the sharpened odd remainder (5.14) by the latter. Formula (1.7) gives the stated limits. In particular the ratios are bounded on $[0, U]$, and (1.8)–(1.9) follow as well. □

5.4 An exact check at small slow parameter

The quadratic coefficient in Proposition 2.2 provides an all-index consistency check. For odd $n \ge 3$,

$$\begin{aligned}\lim_{u\to 0}\frac{A^nC_n(u)}{4u^2b_n} &= \frac{A^n}{4b_n}\frac{n}{4}r_{n-1}\\ &= c\frac{n^2}{(n-1)^2}\zeta(n-1)^2 = c\left\{1 + \frac{2}{n} + O(n^{-2})\right\}.\end{aligned} \tag{5.16}$$

Our expansion gives the same result: $H_1(cu)/u \to c$ and $K_1(u)/u \to 2c$, with a remainder that stays uniform after this division. For even n, the corresponding linear-coefficient identity is

$$\lim_{u\to 0} \frac{-A^n C_n(u)}{4ub_n} = \zeta(n)^2 = 1 + O(2^{-n}).$$

These checks also explain the different powers of u in the two parity sectors. If $u = u_n > 0$ tends to zero at any rate while n tends to infinity, (5.15) gives

$$C_n(u_n) \sim \begin{cases} -4u_n b_n A^{-n}, & n \text{ even}, \\ A u_n^2 b_n A^{-n}, & n \text{ odd}. \end{cases}$$

Thus the late-coefficient theorem remains informative when the slow parameter follows the physical specialization $u = X^{-1}$.

6 Logarithmic and dilogarithmic Borel boundary models

The coefficient theorem gives more information than the Borel radius. It identifies explicit functions carrying the two leading boundary singularities. The regularity statements in this section concern limits from the closed disk. Analytic continuation outside that disk is proved separately in Section 7.

Put $z = cu$ and abbreviate the two combinations in (1.7) by

$$P(u) = du + \frac{1+u}{2} z^2, \qquad Q(u) = (2+u)z. \tag{6.1}$$

Thus $K_\epsilon = PH_\epsilon + QH'_\epsilon$. All logarithms below mean their Taylor branches at $\xi = 0$; the dilogarithms have the same convention. Define

$$B_{0,u}(\xi) = \frac{2u}{\xi} \left\{ e^{-z} \log(1 - \xi/A) + e^{z} \log(1 + \xi/A) \right\}, \tag{6.2}$$

$$B_{1,u}(\xi) = B_{0,u}(\xi) - \frac{2u}{\xi} \left\{ (P-Q)e^{-z} \operatorname{Li}_2(\xi/A) + (P+Q)e^{z} \operatorname{Li}_2(-\xi/A) \right\}. \tag{6.3}$$

The factors ξ^{-1} have removable singularities at zero.

Theorem 6.1 (Boundary models with controlled remainders)**.** *For every $u \in \mathbb{C}$, there are functions $\mathcal{H}_{0,u}$ and $\mathcal{H}_{1,u}$, holomorphic in $|\xi| < A$, such that*

$$\widehat{V}_u = B_{0,u} + \mathcal{H}_{0,u} = B_{1,u} + \mathcal{H}_{1,u}. \tag{6.4}$$

The function $\mathcal{H}_{0,u}$ extends continuously to $|\xi| \le A$. Both $\mathcal{H}_{1,u}$ and its complex derivative extend continuously to that closed disk. These extensions, and the convergent series defining them, are locally uniform in u.

For every $u \neq 0$, both $\xi = A$ and $\xi = -A$ are genuine singularities of $\widehat{V}_u$, whose Taylor radius is exactly A.

Proof. Resolve the hyperbolic functions in Theorem 1.1 into their two exponentials. For all sufficiently large n, uniformly for $|u| \le U$,

$$\begin{aligned} \frac{C_n(u)}{\Gamma(n)} = &- \frac{2u}{nA^n} \left\{ e^{-z} + (-1)^n e^{z} \right\} \\ &- \frac{2u}{n^2 A^n} \left\{ (P-Q)e^{-z} + (-1)^n (P+Q)e^{z} \right\} + O_U\!\left(\frac{|u|}{A^n n^3} \right). \end{aligned} \tag{6.5}$$

For example, at even n the second brace is $2(P\cosh z + Q\sinh z)$; at odd n its negative is $2(P\sinh z + Q\cosh z)$. This verifies the signs without any singularity-transfer inference.

Now use the convergent identities

$$\log(1-w) = -\sum_{n\geq 1}\frac{w^n}{n}, \qquad \mathrm{Li}_2(w) = \sum_{n\geq 1}\frac{w^n}{n^2}, \qquad |w|<1.$$

The first line of (6.5) is precisely the coefficient sequence of (6.2). The first two lines together give (6.3). Consequently the coefficients of $\mathcal{H}_{0,u}$ at ξ^{n-1} are $O_U(|u|A^{-n}n^{-2})$, and those of $\mathcal{H}_{1,u}$ are $O_U(|u|A^{-n}n^{-3})$. The finitely many omitted indices cause no difficulty: their coefficients are entire functions of u.

On $|\xi|\leq A$, the first remainder series is therefore dominated by a constant times $\sum n^{-2}$. The second remainder and its termwise derivative are dominated by $\sum n^{-3}$ and $\sum n^{-2}$, respectively. The Weierstrass criterion proves all the asserted extensions, including local uniformity in the parameter. Equation (6.5) also proves that the original Borel series converges in $|\xi|<A$.

For $0<\varepsilon<1$, (6.2) and continuity of its remainder give

$$\widehat{V}_u(A(1-\varepsilon)) = \frac{2u}{A}e^{-z}\log\varepsilon + O_u(1), \tag{6.6}$$

$$\widehat{V}_u(-A(1-\varepsilon)) = -\frac{2u}{A}e^{z}\log\varepsilon + O_u(1). \tag{6.7}$$

Indeed the opposite logarithm tends to $\log 2$, and replacing $(1-\varepsilon)^{-1}$ by 1 contributes $O(\varepsilon|\log\varepsilon|)$. For $u\neq 0$, neither displayed logarithmic coefficient vanishes. Thus neither boundary point admits a holomorphic extension of the principal germ, and its radius is exactly A. □

This argument remains valid when an individual parity amplitude vanishes at a complex value of u. The two boundary amplitudes are exponentials times u, and therefore never vanish for $u\neq 0$. At $u=0$ the defining equation gives $V=0$, so there is no singularity. The exact positive Borel radius also implies that the ordinary q-series $V(q,u)$ has radius zero for each fixed $u\neq 0$: the root growth of $\Gamma(n)$ converts a positive Borel coefficient growth rate into an unbounded ordinary coefficient growth rate.

The continuously differentiable remainder makes one further radial term explicit. With $\sigma\in\{1,-1\}$, set

$$S_\sigma(u) = \frac{2u}{\sigma A}e^{-\sigma z}.$$

There are finite constants $L_\sigma(u)$ such that

$$\begin{aligned}\widehat{V}_u(\sigma A(1-\varepsilon)) = {} & S_\sigma(u)\log\varepsilon + L_\sigma(u)\\ & + S_\sigma(u)\{1-P(u)+\sigma Q(u)\}\varepsilon\log\varepsilon + O_u(\varepsilon).\end{aligned} \tag{6.8}$$

To see this, use $\mathrm{Li}_2(1-\varepsilon) = \zeta(2) + \varepsilon\log\varepsilon - \varepsilon + O(\varepsilon^2|\log\varepsilon|)$ in (6.3), expand $(1-\varepsilon)^{-1}$, and use the bounded derivative of $\mathcal{H}_{1,u}$. The dilogarithm with opposite argument is holomorphic near -1. This proves (6.8) directly. Neither (6.4) nor (6.8) asserts that the remainder extends holomorphically through the two endpoints.

7 Resurgent continuation and sectorial sums

The preceding boundary models follow from coefficient estimates alone. The arithmetic forward kernel supplies the additional analytic information needed for continuation and

summation. We use a standard nonlinear closure theorem, with its hypotheses stated explicitly.

7.1 The arithmetic kernel and its continuation paths

For a formal series $f(q) = \sum_{n\geq 1} f_n q^n$, our Borel convention is $\mathcal{B}f(\xi) = \sum_{n\geq 1} f_n \xi^{n-1}/\Gamma(n)$, as in (1.10). Let

$$\Omega = A\mathbb{Z}.$$

An Ω-continuable germ is holomorphic at zero and admits analytic continuation along every path starting there whose subsequent points avoid Ω. Initial motion inside the regular disk is understood in this definition. When passing to functions of $x = 1/q$, write

$$\mathcal{R}_\Omega = \mathbb{C} \oplus \mathcal{B}^{-1}\{\Omega\text{-continuable germs}\}.$$

Thus $\mathcal{R}_\Omega$ is a space of formal series in x^{-1}. Constants are separated because our Borel transform acts on the part without constant term.

Proposition 7.1 (A normally convergent logarithmic lattice)**.** *The shifted Borel transform of* (1.3) *is*

$$\widehat{R}(\xi) = -\frac{1}{4} - \frac{2}{\xi}\sum_{m\geq 1} d(m)\log\left(1 - \frac{\xi^2}{A^2m^2}\right). \tag{7.1}$$

Its principal expression is holomorphic on $\mathbb{C} \setminus ((-\infty, -A] \cup [A, \infty))$*. The series continues normally on compact subsets of the path surface avoiding* Ω*. In particular* $R(1/x) \in \mathcal{R}_\Omega$*.*

Proof. For $|\xi| < A$, expansion of each logarithm and the absolutely convergent Dirichlet identity $\sum_{m\geq 1} d(m)m^{-s} = \zeta(s)^2$ give

$$-\frac{2}{\xi}\sum_{m\geq 1} d(m)\log\left(1 - \frac{\xi^2}{A^2m^2}\right) = \sum_{j\geq 1} \frac{4\zeta(2j)^2}{2jA^{2j}}\xi^{2j-1}.$$

These are exactly the coefficients $r_{2j}/\Gamma(2j)$; the constant $-1/4$ supplies r_1.

Fix a bounded continuation path and a compact neighborhood of its lift. For all sufficiently large m, the argument $\xi^2/(A^2m^2)$ stays in a fixed disk of radius less than one, so its logarithm retains the Taylor branch and is $O(m^{-2})$, uniformly on that compact set. The sum of its majorants converges because $\sum d(m)m^{-2} = \zeta(2)^2 < \infty$. Only finitely many logarithms need nontrivial continuation; their branch points are the stated lattice points $\pm Am$. This proves normal continuation. The factor ξ^{-1} is removable at zero on the principal sheet, since each logarithm starts quadratically. □

The role of zero in Ω matters. After a loop around a branch point, one logarithm can change by $2\pi\mathrm{i}$, adding a nonzero multiple of ξ^{-1} to (7.1). Thus zero can be singular on another sheet although the original germ is regular there. Our continuation statement permits this behavior and does not promise passage through zero after a nontrivial loop.

7.2 The nonlinear closure input

We use the following established consequences of Sauzin's convolution estimates [6, Theorems 3 and 5, and their summable counterparts 3′ and 5′]. They are quoted here as analytic input.

Proposition 7.2 (Nonlinear resurgence and summability closure)**.** *Suppose $\Omega \subset \mathbb{C}$ is closed, discrete, contains zero, and is stable under addition. Then $\mathcal{R}_\Omega$ is an algebra with the following properties.*

1. *If $f_1, \dots, f_j \in \mathcal{R}_\Omega$ have zero constant term and H is a convergent power series near the origin in j variables, then $H(f_1, \dots, f_j) \in \mathcal{R}_\Omega$.*
2. *The formal maps $x + \mathcal{R}_\Omega$ form a group under composition. In particular a map $G(x) = x + O(1)$ in this group has a compositional inverse there, and $f \circ G \in \mathcal{R}_\Omega$ whenever $f \in \mathcal{R}_\Omega$.*

The same operations preserve 1-summability in an arc of directions. On sufficiently small common sectors, Borel–Laplace summation commutes with these operations, including compositional inversion.

The topology underlying this statement controls analytic continuation on compact subsets of the path surface. Its estimates justify infinite substitution and inverse series; coefficientwise formal existence by itself would not supply these conclusions. The set $A\mathbb{Z}$ satisfies every hypothesis of the proposition.

Theorem 7.3 (Continuation of the quadratic inverse)**.** *For every fixed $u \in \mathbb{C}$, the series $V(1/x, u)$ belongs to $\mathcal{R}_\Omega$. Hence (1.10) admits analytic continuation along every path starting at the origin and thereafter avoiding $A\mathbb{Z}$.*

Proof. First take $u \neq 0$ and define, as formal series,

$$h(t) = \frac{\sqrt{1 - 2u^2 R(t)} - 1}{u}, \qquad G(x) = x \exp\{-h(1/x)\}. \tag{7.2}$$

The square root has constant term one. It is a convergent function of $R(t)$ near zero, so Proposition 7.2 gives $h(1/x) \in \mathcal{R}_\Omega$. It satisfies the exact formal identities

$$h(t) + \frac{u}{2} h(t)^2 + uR(t) = 0, \qquad h(t) = \frac{u}{4} t + O(t^2). \tag{7.3}$$

Analytic substitution also puts $e^{-h(1/x)} - 1$ in $\mathcal{R}_\Omega$, with zero constant term.

We check explicitly the multiplication by x in (7.2). If $f(x) = \sum_{n \geq 1} f_n x^{-n}$ has Borel transform $\widehat{f}$, then

$$\mathcal{B}\{x f(x) - f_1\} = \partial_\xi \widehat{f}(\xi).$$

Differentiation preserves analytic continuation along the allowed paths. It follows that $G(x) = x - u/4 + O(x^{-1})$ belongs to $x + \mathcal{R}_\Omega$. The closure theorem therefore provides its unique compositional inverse $T = G^{-1} \in x + \mathcal{R}_\Omega$.

Set

$$t(q) = \frac{1}{T(1/q)}, \qquad W(q) = h(t(q)). \tag{7.4}$$

Composition closure gives $W(1/x) = h(1/T(x)) \in \mathcal{R}_\Omega$. The identity $G(T(x)) = x$ becomes $q = te^{h(t)}$, so $t = qe^{-W}$. Equation (7.3) now implies

$$W + \frac{u}{2} W^2 + uR(qe^{-W}) = 0.$$

Formal uniqueness in Proposition 2.1 gives $W = V$. For $u = 0$, equation (1.4) gives $V = 0$ directly. □

This is a continuation theorem with an explicit allowed set. It does not assert that every nonzero point of $A\mathbb{Z}$ is singular on every sheet. The genuine singularities at $\pm A$ on the principal boundary were proved independently in Theorem 6.1. No assertion about an all-orders alien derivative or a complete local singularity type is used.

7.3 Growth away from the real directions and analytic inversion

Lemma 7.4 (Growth of the forward kernel on nonreal rays)**.** *For every $\delta \in (0, \pi/2)$ there is $M_\delta > 0$ such that the principal kernel in (7.1) satisfies*

$$|\widehat{R}(\xi)| \le M_\delta\{1 + \log^2(2 + |\xi|)\} \tag{7.5}$$

when $\delta \le \arg\xi \le \pi - \delta$ or $-\pi + \delta \le \arg\xi \le -\delta$.

Proof. The elementary divisor estimate

$$D(M) := \sum_{m \le M} d(m) = \sum_{a \le M} \left\lfloor \frac{M}{a} \right\rfloor \le M(1 + \log M)$$

and summation by parts give $\sum_{m>M} d(m)m^{-2} = O((1 + \log M)/M)$. Put $\rho = |\xi|/A$ and initially suppose $\rho \ge 1$. Split the logarithmic lattice at $m = 2\rho$.

For $m > 2\rho$, Taylor bounds give $|\log(1 - \xi^2/(A^2m^2))| \le C\rho^2/m^2$, so the total tail before division by ξ is $O(\rho \log(2\rho))$. For $m \le 2\rho$, the angular restriction implies

$$|1 - se^{2\mathrm{i}\arg\xi}| \ge \sin\delta\,(1 + s) \qquad (s \ge 0).$$

Together with the upper bound $1 + s$ and the bounded argument of the principal logarithm, this bounds each logarithm by $C_\delta + 2\log(1 + \rho/m)$, hence by $O_\delta(\log(2 + \rho))$. The estimate for $D(2\rho)$ bounds the head by $O_\delta(\rho \log^2(2 + \rho))$. Division by $|\xi| = A\rho$ proves (7.5). The remaining bounded part of each closed angular sector is controlled by holomorphy and the removable value at zero. □

Theorem 7.5 (Sectorial sums satisfy the inverse equation)**.** *Fix $u \in \mathbb{C}$. The series $V(q, u)$ is 1-summable in every direction $\theta \notin \pi\mathbb{Z}$. Let*

$$R_\theta(t) = \int_0^{e^{\mathrm{i}\theta}\infty} e^{-\xi/t}\widehat{R}(\xi)\,\mathrm{d}\xi$$

on its Laplace sector, and write $V_\theta = \mathcal{S}_\theta V$ for the corresponding sum of the inverse series. For sufficiently small q in any smaller common sector,

$$V_\theta(q, u) + \frac{u}{2}V_\theta(q, u)^2 + uR_\theta(qe^{-V_\theta(q,u)}) = 0. \tag{7.6}$$

This is the unique small solution with $V_\theta(q, u) = O(q)$.

Proof. Choose an open arc about θ whose closure avoids the real directions. Proposition 7.1 and Lemma 7.4 give the analytic continuation and exponential growth condition for 1-summability of R throughout this arc. The logarithmic bound is stronger than the required bound of exponential type.

Apply the summable version of Proposition 7.2 to the same operations as in (7.2)–(7.4). Multiplication by x causes no difficulty: its Borel action is differentiation, and Cauchy's estimate on a slightly smaller angular domain preserves an exponential growth bound. Thus h,

$G - x$, $T - x$, and V are 1-summable. Summation commutes with their analytic substitutions and inverse relations on smaller sectors. In particular, putting

$$h_\theta(t) = \frac{\sqrt{1 - 2u^2 R_\theta(t)} - 1}{u}$$

with the branch tending to zero, the summed parametrization has $V_\theta = h_\theta(t)$ and $t = qe^{-V_\theta}$. This proves (7.6). At $u = 0$, all statements follow from $V = 0$.

For uniqueness, the summed forward function and its derivative satisfy $R_\theta(t) = -t/4 + O(t^2)$ and $R'_\theta(t) = -1/4 + O(t)$ on every smaller sector. The map

$$w \longmapsto -\frac{u}{2}w^2 - uR_\theta(qe^{-w})$$

therefore has derivative $O_u(|q|)$ on a disk $|w| \le M|q|$ and maps that disk into itself for a suitable fixed M and sufficiently small q. The contraction argument proves uniqueness there. Any solution that is $O(q)$ enters such a disk after decreasing the sector radius. □

The preceding sum is defined by Borel–Laplace summation of the divergent formal tail. It is not an ordinary sum of its exponential sectors. The real-direction completions of the forward Lambert function are specified in [1]; their upper and lower choices can subsequently be transported through (7.6). The continuation theorem does not require identifying those choices with an arithmetic average.

8 Compatibility with the exact Stokes transport

The factor $e^{-Au/4}$ in the positive-action boundary amplitude has an independent analytic interpretation. It is the finite change in the exponential $e^{-A/t}$ when the inverse coordinate changes from q to $t = qe^{-V}$. We make that statement precise using the exact forward completion already established in the incoming report.

8.1 The exact forward input and the sign convention

For positive t define the exact remainder

$$R_{\mathrm{ex}}(t) = \mathcal{F}(t) - \frac{1}{2}\{\log(1/t) + \gamma\}^2 - C_*. \tag{8.1}$$

It includes the linear term $-t/4$ and has the formal expansion $R(t)$. This convention differs from the predecessor's calligraphic $\mathcal{R}_0$, which begins at degree two. The two lateral sums of our entire formal remainder satisfy

$$R_\pm(t) = R_{\mathrm{ex}}(t) \pm \frac{1}{2}J(t), \qquad J(t) = 4\pi\mathrm{i}\sum_{m\ge1} d(m)E_1(Am/t), \tag{8.2}$$

where

$$E_1(x) = \int_x^\infty \frac{e^{-v}}{v}\,\mathrm{d}v, \qquad x > 0.$$

The exact identification of the arithmetic mean with (8.1) is the Mellin–Voronoi completion theorem of [1, Section 5]; it is an imported analytic identity. The sign of its jump also follows directly from (7.1): upper continuation of the first logarithm past a positive branch point gives imaginary part $-\pi$. Thus upper minus lower continuation of $\widehat{R}$ is positive $4\pi\mathrm{i}/\xi$ times the accumulated divisor weight, and its Laplace transform is $J(t)$.

The series for J converges normally for $\mathrm{Re}(1/t) > 0$. For $t \to 0^+$,

$$J(t) = \frac{4\pi \mathrm{i}}{A}\, te^{-A/t}\{1 + O(t)\}, \tag{8.3}$$

$$J'(t) = \frac{4\pi \mathrm{i}}{t} \sum_{m\geq 1} d(m) e^{-Am/t}. \tag{8.4}$$

The first formula follows by one integration by parts in E_1, with the terms $m \geq 2$ exponentially smaller. Termwise differentiation proves the second. These formulas also hold with uniform bounds in the small complex neighborhoods used below.

8.2 Uniform transport in the slow parameter

Proposition 8.1 (Leading inverse jump with its shifted amplitude)**.** *Fix $U > 0$. For all sufficiently small real $q > 0$, uniformly for $0 < u \leq U$, let V_η be the small solution of*

$$V_\eta + \frac{u}{2} V_\eta^2 + uR_{\mathrm{ex}}(qe^{-V_\eta}) + \frac{\eta u}{2} J(qe^{-V_\eta}) = 0, \qquad \eta \in \{0, 1, -1\}. \tag{8.5}$$

Put $t_0 = qe^{-V_0}$ and

$$D = 1 + uV_0 - ut_0 R'_{\mathrm{ex}}(t_0).$$

Then

$$V_+ - V_- = -\frac{uJ(t_0)}{D} + O_U(u^2 qe^{-2A/q}), \tag{8.6}$$

$$= -\frac{4\pi \mathrm{i}}{A}\, uq \exp\{-A/q - Au/4\}\{1 + O_U(q)\}. \tag{8.7}$$

Consequently, under the physical specialization (1.5), the lateral inverses $t_\pm = qe^{-V_\pm}$ satisfy

$$t_+ - t_- = \frac{\mathrm{i}}{\pi} \frac{q^2}{X} \exp\left\{-\frac{A}{q} - \frac{A}{4X}\right\}\{1 + O(q)\}, \qquad X \to +\infty. \tag{8.8}$$

Proof. The finite Mellin expansion, with differentiation on a smaller sector, gives

$$R_{\mathrm{ex}}(t) = -t/4 + O(t^2), \quad R'_{\mathrm{ex}}(t) = -1/4 + O(t), \quad R''_{\mathrm{ex}}(t) = O(1).$$

For example the differentiated bounds follow either by shifting (2.1) farther before differentiating, or by Cauchy's estimate applied to the finite-order remainder. They are uniform near the positive ray. A contraction on $|v| \leq M_U uq$ therefore gives the solution at $\eta = 0$ and

$$V_0 = uq/4 + O_U(uq^2), \qquad D = 1 + O_U(q). \tag{8.9}$$

The derivative of the fixed-point map is $O_U(q)$ on this disk. The exponentially small added term in (8.5) preserves the contraction for $|\eta| \leq 1$ and gives

$$\delta_\eta := V_\eta - V_0 = O_U(uqe^{-A/q}).$$

Here are bounds sufficient to retain the square of the exponential in the remainder. In a complex disk $|v - V_0| \leq c_0 q$, with c_0 fixed,

$$|\exp\{-Ae^v/q\}| \leq C_U e^{-A/q}.$$

Indeed $V_0 = O_U(q)$ and $e^v - 1 = O_U(q)$. The divisor series and (8.4) then give

$$|J(qe^{-v})| \leq C_U qe^{-A/q}, \qquad |\partial_v J(qe^{-v})| \leq C_U e^{-A/q}.$$

The second derivative of $v + uv^2/2 + uR_{\mathrm{ex}}(qe^{-v})$ is $O_U(u)$. Taylor expansion at V_0 in (8.5) consequently yields

$$D\delta_\eta + \frac{\eta u}{2} J(t_0) = O_U\Big(u|\delta_\eta|^2 + ue^{-A/q}|\delta_\eta|\Big) = O_U(u^2 qe^{-2A/q}).$$

Subtracting the equations for $\eta = 1$ and -1 proves (8.6). The bounds retain all displayed powers of u, so they are uniform down to $u = 0$.

Finally (8.9) gives

$$t_0 = q\{1 - uq/4 + O_U(uq^2)\}, \qquad \frac{A}{t_0} = \frac{A}{q} + \frac{Au}{4} + O_U(q).$$

Insert these relations and (8.3) into (8.6) to obtain (8.7). The Taylor expansion of qe^{-v} gives $t_+ - t_- = -t_0(V_+ - V_-)$ to the same relative accuracy. Since $4\pi/A = 1/\pi$ and $u = 1/X$, this proves (8.8). □

The positive Borel logarithm in (6.2) has coefficient $2ue^{-Au/4}/\xi$. Its upper-minus-lower logarithmic discontinuity has the sign $-2\pi\mathrm{i}$, consistent with (8.7). This is a consistency check between two proved calculations. We have not inferred a complete Stokes discontinuity from a merely continuous boundary remainder. Nor do we identify an arithmetic mean after nonlinear inversion with a nonlinear median summation operation. The exact inverse transport in [1, Section 8] already distinguishes those operations.

9 Reproducible verification and numerical behavior

The proofs establish the infinite statements. The accompanying program checks exact finite identities, evaluates complete finite coefficient formulas, and tests the scale of the asymptotic errors. It does not use the proposed asymptotic formulas to generate the coefficients being compared.

9.1 Exact checks

The script `verification/verify.py` constructs the polynomials $C_1, \dots, C_{12}$ over $\mathbb{Q}[u]$ by two independent routes: coefficient recursion in the implicit equation and the finite Lagrange formula (2.5). It then substitutes the result back into (1.4) through degree twelve. It checks the sign, valuation, degree, Catalan top coefficient, and the exact lowest-coefficient identities of Proposition 2.2.

The same run checks the rational factorial inequalities through $n = 100$. For the gamma ratio it checks the rational consequence $n^{k-1}b_{n-k+1}/b_n \le k3^k$ for every $1 \le k \le n$, including $k = n$; the sharper ke^k bound is proved analytically in Lemma 3.1. Convolution powers are checked through $n = 32$. Exact symbolic algebra checks the first-correction sums. In total, the executed run passed **5,890 exact assertions**. The JSON record separates those assertions by category; `coefficients_exact.txt` contains the complete degree-twelve coefficient list.

9.2 Complete finite sums and independent precision checks

Direct factorials become very large. The numerical implementation therefore normalizes the convolution recurrence by its endpoint gamma factor before forming (3.3). The main table evaluates *every* finite summand $1 \le k \le n$: there is no mathematical truncation in these coefficient values.

The recorded grid contains 70 cases, with

$$n \in \{64, 65, 128, 129, 256, 257, 512, 513, 1024, 1025\},$$

and the seven parameter choices $u = 0.01, 0.05, 0.1, 0.2, 0.5$, $u = 0.1 + 0.05\mathrm{i}$, and $u = 1/\log(n+e)$. The last choice exercises a parameter tending to zero. The main arithmetic uses a 64-bit-significand extended-range `longdouble`, whose properties are recorded and checked at runtime. A platform that aliases this type to ordinary binary64 is rejected explicitly.

An independent 80-decimal-digit `mpmath` calculation at the recorded cross-check indices agrees with the main calculation to within 2.36×10^{-19} in the program's relative-or-absolute normalization. Evaluation of the exact coefficient polynomials agrees with the high-precision route to within 4.68×10^{-81}. A separate binary64 convolution computation at larger indices differs by at most 3.12×10^{-16}. That secondary check truncates at $k = 256$; its omitted contribution to $D_n/(4b_n)$ is bounded by the exact rational certificate recorded in `results.json`, whose value is less than 2.861×10^{-18} for $|u| \le 1/2$. This certificate bounds omitted terms, not floating-point roundoff. The main table needs no such truncation certificate.

For the table and plots define

$$Z_n = \frac{(-1)^{n+1} A^n C_n}{4ub_n}, \qquad E_0 = \left| \frac{Z_n - H_\epsilon}{H_\epsilon} \right|, \qquad E_1 = \left| \frac{Z_n - H_\epsilon - K_\epsilon/n}{H_\epsilon} \right|, \quad \epsilon = n \bmod 2,$$

where $H_\epsilon = H_\epsilon(cu)$. Table 1 gives representative errors at $u = 0.1$. The normalization is the leading amplitude H_ϵ, not the computed coefficient. Figure 1 shows both parities and the scaled second-order residual.

Table 1: Leading-amplitude-normalized errors for even coefficients at $u = 0.1$. Every finite Lagrange summand is included.

n	Leading amplitude only: E_0	With the $1/n$ correction: E_1
64	4.52796×10^{-2}	3.96799×10^{-3}
128	2.15988×10^{-2}	9.42970×10^{-4}
256	1.05580×10^{-2}	2.30119×10^{-4}
512	5.22080×10^{-3}	5.68550×10^{-5}
1024	2.59610×10^{-3}	1.41311×10^{-5}

At the largest even and odd indices, the first correction reduces the observed errors by factors between 37 and 592 over the tested parameters. The recorded software versions, precision checks, full CSV, and machine-readable results make this comparison reproducible. No interval certification of the floating-point output is claimed.

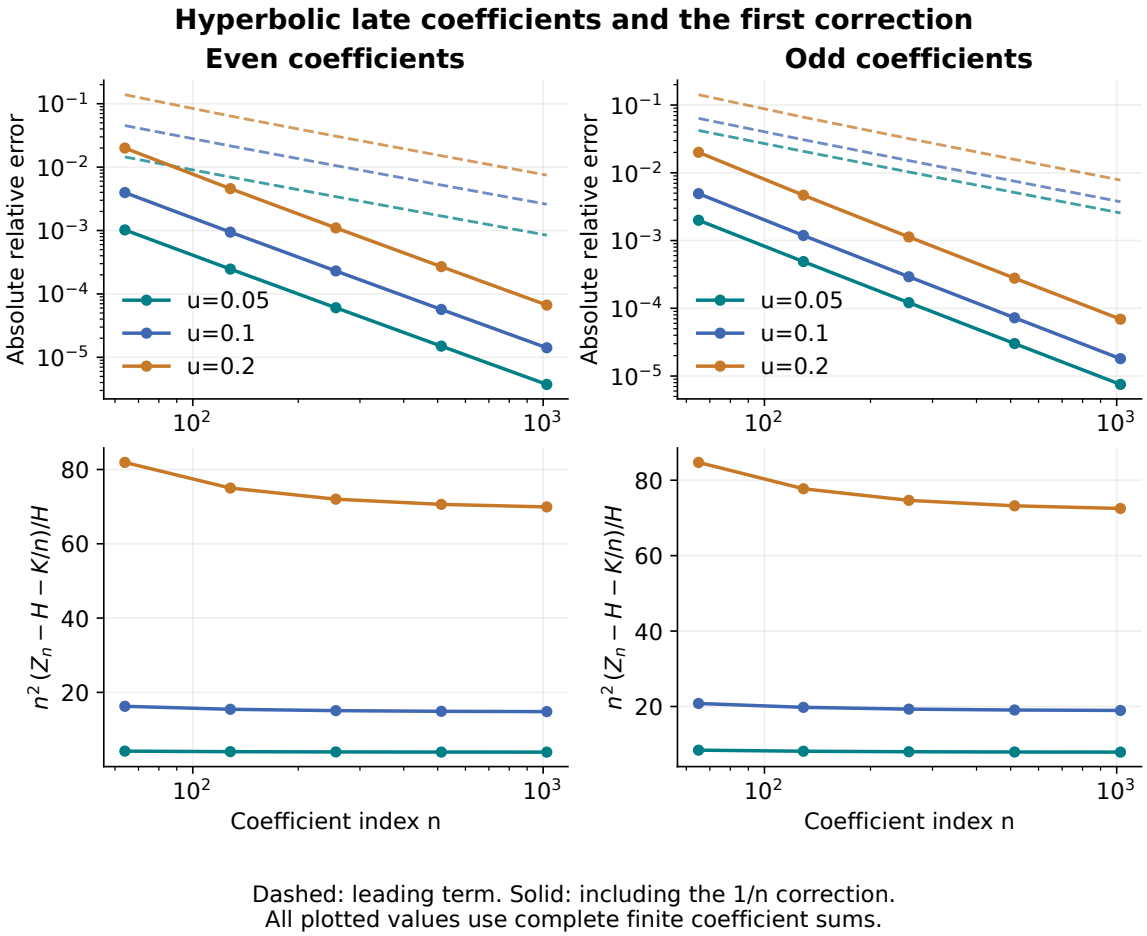


Figure 1: Executed finite-sum diagnostics from the supplied program. The observed first- and second-order decay agrees with the proved uniform expansion. The plot is evidence of implementation consistency, not a proof of an asymptotic estimate.

10 Source audit and proposed repository integration

10.1 The inspected snapshot and the limits of the audit

All repository references in this contribution are pinned to commit

`8f5afd4d7565df808359f6a0c22f87f79a119bb9`

dated 12 October 2026. The research used the Polylogarithms and Transseries indexes, relevant source chapters, and the three incoming research archives listed below. The package's `provenance.json` records exact paths and cryptographic hashes.

1. `Resonant_Lambert_Inverse_Transseries_2026-10-12.zip`: the principal predecessor, including its inverse-coefficient theorem, exact completion and Stokes transport, and explicit research questions.
2. `ProveIt_Lambert_Polylog_Order_Jets_2026-10-12.zip`: order derivatives, resonant jets, and the arithmetic Lambert interface.
3. `ProveIt_Quadratic_Polylog_Boundaries_2026-10-12.zip`: the quadratic feedback problem and its boundary regimes.

The related integrated chapters are `01-resonant-jets.tex`, `01-lambert-moments.tex`, and `01-quadratic-feedback.tex` under the Polylogarithms manuscript. The related Transseries reports include `Polylogarithm_Resonance_And_Inverse_Transseries`, `Quadratic_Polylogarithm_Feedback_Boundary_Completion`, and `Resonant_Polylogarithm_Transseries`. In particular, the last report already proves an arbitrary-order Fermi–Dirac Borel completion. We do not present that existing result as a new contribution.

No specific false theorem was established in this scoped audit. The unproved hyperbolic amplitudes are explicitly labelled conjectural in their source; its coarser coefficient bound is valid for the purpose there. Resolving that conjecture is an addition to the source, not a retroactive claim that its proof was incorrect. This audit does not certify every file in either research program.

10.2 Concrete updates supported by the present proofs

The following integration changes are justified without expanding the claims beyond what is proved.

1. **Promote the two amplitude conjectures.** In Section 11.1 of the principal predecessor, replace the conjectural status of `eq:conj-even` and `eq:conj-odd` by a reference to Theorem 1.1. State uniformity on $0 < u \le U$ and preserve the distinction between coefficient index and physical target size. The first correction can be added as a separate strengthening.
2. **Sharpen the inverse Borel conclusion.** The source's radius argument is strengthened by Theorem 6.1: both nearest real actions are genuine singularities for every fixed complex $u \neq 0$. Include the explicit logarithmic and logarithmic–dilogarithmic models, with their stated boundary regularity.
3. **Resolve the continuation part of Section 11.2.** Cite Theorem 7.3 and the original closure theorem of Sauzin. Formulate the allowed set as $A\mathbb{Z}$, including the possibility

of a singular origin on another sheet. Leave the exact active lattice and full local or alien description as research questions.

4. **Retain the formal and analytic distinctions.** A zero-radius sector series is not made convergent by inserting $u = 1/X$. The sectorial inverse is provided by summation as in Theorem 7.5. Also retain the explicit $-t/4$ term when translating the source's degree-two remainder to the convention of this article; omitting it removes the amplitude shift.

The accompanying `INTEGRATION.md` gives suggested destination paths, stable theorem labels, and reviewable replacement prose. It proposes a new report directory under `Analysis/Transseries` and a cross-reference from Polylogarithms. No existing source file is silently replaced, and this package does not contain a repository push.

11 Further research questions

The proofs isolate several concrete next steps. We separate problems that appear approachable by extending the present coefficient argument from those requiring additional analytic machinery.

11.1 All inverse-index orders and higher boundary regularity

Can one prove, for each fixed N, a locally uniform expansion

$$\frac{(-1)^{n+1}A^nC_n(u)}{4ub_n} = \sum_{j=0}^{N-1} \frac{K_{\epsilon,j}(u)}{n^j} + O_{U,N}(n^{-N}), \qquad \epsilon = n \bmod 2,$$

with $K_{\epsilon,0} = H_\epsilon(cu)$ and $K_{\epsilon,1} = K_\epsilon(u)$? The expected route is to retain all configurations with a fixed small total excess away from the large index, expand the gamma ratios and $T_{n,k}$ to the same order, and bound the remainder uniformly before summing over k. An explicit recursion for the resulting polynomial coefficients multiplying $\cosh(cu)$ and $\sinh(cu)$ would be more useful than isolated higher-order formulas.

Such a theorem would produce Borel models built from $\mathrm{Li}_1, \dots, \mathrm{Li}_N$, where $\mathrm{Li}_1(w) = -\log(1-w)$, with a C^{N-1} remainder on the closed disk. More precisely, keeping the terms $j = 0, \dots, N-1$ above leaves Borel coefficients $O(A^{-n}n^{-N-1})$, whose derivatives through order $N-1$ converge absolutely on the boundary. Smooth boundary control of every finite order would still not, by itself, identify a complete local analytic singularity type.

11.2 A paired-action factorial calculus

The finite-phase theorem proves the leading transformation law

$$\kappa_\omega \longmapsto \kappa_\omega e^{\lambda_\omega \rho u}, \qquad \lambda_\omega = A/\omega.$$

Can the factorial asymptotic calculus of [5] be extended in a form that handles several equal-modulus actions and cancellations uniformly in a parameter? For the present model one seeks simultaneous coefficient expansions in $\Gamma(n-1-j)A^{-(n-1-j)}$ and $\Gamma(n-1-j)(-A)^{-(n-1-j)}$, with a rigorously justified composition rule. That result should reproduce Theorem 1.1 and explain which finite forward coefficients control each successive order. It would also apply to root-of-unity twists and character-weighted Lambert sums.

11.3 Which arithmetic lattice points are active?

Theorem 7.3 confines possible singularities to $A\mathbb{Z}$, and Theorem 6.1 proves activity at $\pm A$. Determine the singular behavior at $\pm 2A, \pm 3A, \dots$ on specified continuation paths. At $2A$, for example, the forward divisor weight $d(2)$ can interact with two nonlinear contributions of action A. The problem is to separate these sources without double counting and to decide whether any cancellations occur for special u. A precise result should specify the sheet, path convention, and local coordinate at every action.

11.4 The full local singularity and Stokes operator

The continuous and continuously differentiable boundary remainders do not determine a complete analytic singularity type. Can one show that the primitive-action singularities admit a logarithm times a convergent or explicitly resurgent germ, and compute that germ to all orders? The exact analytic transport in Section 8 gives a strong constraint but is not a substitute for a local Borel theorem. Care is needed at the origin on other sheets: differentiation in the coordinate-inversion proof can change the order of a pole. A global claim of *simple resurgence* needs a separate proof.

11.5 A growing slow parameter and a new saddle regime

The present bounds are uniform on every fixed disk in u, with no lower bound. They are not uniform as $|u| \to \infty$. The leading exponential sum favors k of size approximately $|cu|$. When $u = u_n$ grows with n, both the gamma-ratio correction k^2/n and the feedback correction uk^2/n can become significant. Identify the largest growing ranges that still admit relative hyperbolic asymptotics, and determine the transition law when the feedback correction is order one. The size of the relative correction $K_\epsilon(u)/(nH_\epsilon(cu))$ suggests a positive-real transition near u of order $n^{1/3}$; this is a scaling heuristic, not a theorem.

11.6 Optimal truncation for the physical inverse

For fixed u, the ratio of successive coefficient magnitudes suggests optimal sector truncation near $n = A/|q|$, modulo the parity amplitudes. The physical problem couples $q = e^{\gamma - X}$ and $u = 1/X$, placing that truncation order far beyond polynomial size in X. Can one obtain a sharp remainder for the real inverse after optimal truncation, together with a terminant description of the lateral jump? The uniform coefficient theorem supplies necessary late-term data; such a remainder theorem also needs analytic estimates on the summed inverse and its Stokes neighborhoods.

11.7 Moving away from exact resonance

The incoming work contains parameter jets at exact polylogarithmic resonance. Extend the present result to nonzero order displacement ε, with coefficient bounds uniform as $\varepsilon \log(1/q)$ changes size. The coalescing-pole core depends on this combined variable, and its order jets contain logarithms; the regular forward coefficients also vary with ε. In the source's normalization, for $n \geq 2$ and ε near zero,

$$r_n(\varepsilon) = \frac{2\{1 + (-1)^n \cos(\pi\varepsilon)\}}{A^n n!} \Gamma(n-\varepsilon)\Gamma(n+\varepsilon)\zeta(n-\varepsilon)\zeta(n+\varepsilon).$$

Thus an odd regular tail is already activated away from resonance, as proved in [1]. The new target is a uniform theorem for the late *inverse* coefficients, specifying when this mechanism and nonlinear odd-sector production reinforce or cancel one another, with the corresponding action amplitudes.

11.8 Exact verification and reusable formal components

The finite coefficient formula, valuation identities, and the reciprocal-binomial convolution bounds are natural candidates for proof-assistant formalization. They involve finite algebra and inequalities before any analytic continuation is introduced. A separate numerical project could replace the diagnostic arithmetic by interval bounds for selected large-index coefficients and for the sectorial inverse. Neither task should be confused with the other: finite exact computations check an implementation, whereas a formalized uniform convolution theorem would certify the step that turns all finite coefficients into an asymptotic result.

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